



Create precise variable names

Python has certain spelling or grammatical rules to follow just like any other language.

In programming, we call these rules naming conventions and naming restrictions.

Naming conventions are consistent guidelines that describe the content, creation date, and version of a file in its name. Naming restrictions are rules built into the syntax of the language itself that must be followed. In Python, there are some important naming conventions to keep in mind. One of these is to avoid keywords. A keyword is a special word that is reserved for a specific purpose and that can only be used for that purpose.

You've already encountered some keywords such as `for`, `in`, `if`, and `else`. There are others too, which you'll learn about soon. Keywords should never be used when naming variables. Thankfully, you don't have to worry about accidentally using a keyword. First of all, keywords will appear in a special color in most coding environments. So a good rule to follow is that if you're naming a variable and the name changes color, don't use that name. If you're thinking, but I like colorful variable names and you try to assign a value to a keyword anyway, the computer will say not so fast and throw an error.

Python will let you make spelling mistakes, but it won't let you use a keyword for a variable name. By the way, throw an error is a common phrase that every coder knows

only too well. It just means that the computer returns an error and your code doesn't successfully run. Also, some names are reserved for existing functions. For example, `print` and `str`. So you should also avoid using names of existing functions. To sum up, a big thing to keep in mind regarding variable naming conventions is that you don't want to use reserved keywords or functions.

Precision is essential in programming. This is why there are naming restrictions for variables. For instance, variables can only include letters, numbers, and underscores. This means you can't use spaces, tabs, or special characters, such as the dollar sign or ampersand. Another rule to keep in mind is that while variable names can contain numbers, they must start with a letter or underscore. Also, Python is case sensitive, which means capitalization matters. Lastly, variable names cannot include parentheses. This is because parentheses have other uses in Python. Let's review some examples of effective and ineffective variable names. Any underscore a underscore variable is a valid variable name. Any underscore a underscore variable underscore two is also good. However, one underscore is underscore a underscore number would be invalid because variable names must start with a letter or underscore. Apple's underscore ampersand underscore oranges is also invalid because it uses the special character ampersand. You do have some flexibility when naming your variables.

Since these are references you create, these conventions and restrictions just help make them consistent and useful. Now, let's go back to parentheses to understand more about how these function in Python. When doing calculations, the rules for parentheses follow the mathematical order of operations. For instance, if we input two times three plus four in parentheses, Python will read three plus four first because it's

following the order of operations. This is equal to 14. But two times three in parentheses plus four equals 10. That's because the operation within parentheses will always be completed first.

By the way, if we don't use any parentheses, Python will follow the standard mathematical order of operations. Naming conventions and restrictions for variables help maintain consistency and usefulness as you continue to use Python for a variety of activities. A key part of working with Python as a data analytics professional is being able to effectively name your variables to create meaningful code. Soon, we'll explore conversions and data types. Bye for now.

Naming conventions

Consistent guidelines that describe the content, creation date, and version of a file in its name

Naming restrictions

Rules built into the syntax of the language itself that must be followed

Keyword

A special word that is reserved for a specific purpose and that can only be used for that purpose

Naming restrictions for variables

- Only include letters, numbers, and underscores
- Must start with a letter or underscore
- Case-sensitive
- Cannot include parentheses

Example variable names

any_a_variable ✓

any_a_variable_2 ✓

1_is_a_number ✗

Apples_&_oranges ✗